Software Requirements Specification

for

AI Prompt Engineering Learning Hub

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1. Introduction:

Project Title: AI Prompt Engineering Learning Hub

1.1 Project Overview

The AI Prompt Engineering Learning Hub is an interactive, web-based educational platform designed to teach foundational and advanced industrial prompt engineering methodologies. Traditional learning methods (static blogs, videos) lack real-time feedback, leaving users in a trial-and-error cycle. This platform bridges the gap by embedding an LLM abstraction framework that acts as a real-time "Prompt Critic," evaluating input quality, scoring structural integrity, and suggesting optimizations to help users transition from casual natural language interactions to precise, programmatic prompting.

1.2 Objectives

- **Architect an Interactive Hub:** Build a comprehensive web application dedicated to practical, hands-on instructional pipelines teaching foundational and advanced industrial prompt methodologies.
- **Deploy an Automated Prompt Evaluation Framework:** Embed cloud LLM APIs to provide objective grading metrics, semantic error breakdowns, and contextual structural recommendations on user-crafted inputs.
- **Establish Domain-Specific Sandbox Ecosystems:** Deliver isolated test execution sandboxes packaged with pre-built schema variations for diverse technical domains including code compilation, logic debugging, and data matrix extraction.
- **Gamify Educational Progression:** Engineer user-retention mechanisms incorporating structural milestones, timed engineering challenges, XP accrual, and skill-badge awards to sustain engagement.

1.3 Scope

The scope encompasses a complete web-based software infrastructure featuring secure authentication, a user management dashboard, step-by-step practical modules (covering Zero-Shot, Few-Shot, Role-play, and Chain-of-Thought workflows), an AI prompt playground, a marketplace, and a code compiler export utility. The system will allow users to learn prompt engineering concepts, test prompts, receive AI-generated evaluations, participate in challenges, and access a community prompt library. Administrators will manage users, learning content, and platform activities.

2. Functional Requirements

2.1 User Authentication

FR-01: Users can register an account.

FR-02: Users can log in using email and password.

FR-03: Users can log in using Google OAuth.

FR-04: Users can reset forgotten passwords.

FR-05: Users can log out securely.

2.2 User Dashboard

FR-06: Users can view learning progress.

FR-07: Users can view completed modules.

FR-08: Users can track XP points and badges.

FR-09: Users can access bookmarked prompts.

2.3 Learning Modules

FR-10: Users can access prompt engineering lessons.

FR-11: Users can complete practical exercises.

FR-12: Users can view examples of prompt techniques.

FR-13: Users can take quizzes and assessments.

2.4 AI Prompt Playground

FR-14: Users can submit prompts for evaluation.

FR-15: AI analyzes prompt quality.

FR-16: System generates prompt scores (0–100).

FR-17: AI provides improvement suggestions.

FR-18: AI generates optimized prompt versions.

2.5 Reverse Prompt Engineering Challenges

FR-19: Users can participate in prompt-solving challenges.

FR-20: System evaluates user solutions.

FR-21: Users receive challenge scores and feedback.

2.6 Social Marketplace & Export Tools

FR-22: Users can tag, categorize, publish, search, and upvote enterprise prompt definitions along with standard conversation trees in a public marketplace.

FR-23: The system will provide a one-click macro copy/export tool to extract raw code formats for external programmatic execution loops.

3. Non-Functional Requirements

NFR-01: **Responsive UI:** The frontend must utilize a responsive design engine built with React.js and Tailwind CSS for cross-device readability.

NFR-02: **Fast API Response:** The backend logic layer (Node.js/Express.js REST Ecosystem) must optimize data pipelines to ensure low-latency communication between user dashboards and cloud microservices.

NFR-03: **Secure Authentication Engine:** User data states, token handshakes, and access boundaries must be verified using strict JWT or Firebase security protocols.

NFR-04: **Data Integrity & Management:** Relational or document-oriented schemas (MongoDB / PostgreSQL) must secure and properly index user progress records, high scores, and script archives.

NFR-05: **Robust AI Pipelines:** Integration with external Google Gemini/OpenAI API SDKs must securely pipe contextual boundaries and prevent generic text wrapping failure modes.

NFR-06: **Usability:** User interface should be simple and intuitive. Learning modules should be easy to navigate.

NFR-07: **Reliability:** System availability should be at least 99%. User progress data should be stored reliably.

NFR-08: **Scalability:** The system should support future feature expansion. Cloud deployment should allow increased user capacity.

NFR-09: **Compatibility:** The platform should work on desktop, tablet, and mobile devices. It should support modern web browsers.

4. User Roles

Based on the architectural specifications outlined in the proposal, the system establishes the following key user roles:

- **Registered User (Student / Technical User):** Regular platform users who interact with dashboards, complete modules, accrues XP, tests strings in the sandbox, and downloads/publishes configurations in the marketplace.
- **AI Prompt Critic (System/Automated Role):** The native embedded LLM abstraction abstraction framework that acts as an automated auditor, mathematically grading parameters and synthesizing semantic corrections.
- **Administrator (Implicit System Owner):** Manages backend platform microservices, system availability, curriculum indices, and marketplace infrastructure.
- **Guest User:** View basic platform information. Browse selected public prompts. Register for an account.

5. Use cases

5.1 Use Case 1: Real-time Prompt Evaluation

- **User Action:** A student enters a custom prompt sequence into the AI Prompt Playground sandbox and hits "Evaluate".
- System Response: The system routes the text stream through the LLM abstraction framework. It processes contextual boundary metrics, calculates a composite quality score (0-100), outputs semantic error breakdowns, and dynamically renders alternative optimized prompt structures on the interface.

5.2 Use Case 2: Reverse Prompt Engineering Challenge

- **User Action:** A user opens a gamified lab module, reads a target object output structure, writes an input string to replicate it, and submits.
- **System Response:** The system compares the result generated by the user's prompt against the target state using comparative semantic metrics. If successful, it triggers a milestone flag, increments the user's XP score, and awards the corresponding skill-badge on their dashboard.

5.3 Use Case 3: Prompt Exploration and Exporting

- **User Action:** A user searches the Global Prompt Marketplace, filters by technical domains, upvotes an entry, and clicks the macro export button.
- **System Response:** The platform indexes database records to surface matching configurations. Upon export activation, it builds a one-click macro format payload ready for programmatic execution loops across remote AI workflows.